SAT Math Mastery · Session 1

Quadratics and Nonlinear Functions

Time	3 hours: Part A in class (140 min solving) + 40 min self-check and error log. Part B is the self-assessment, done independently.
Topics	parabolas: vertex, zeros and intercepts · function notation and composition · transformations · exponential and power functions · direct, inverse and joint variation · equivalent forms · undefined values
Name / Date	_____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos, one expression per line; tap a grey dot to read an intersection, intercept or vertex. A letter such as a, c or k becomes a slider when you accept Desmos's offer to add one.

1. HARD · DESMOS

The function g satisfies $g(kn) = n + 10$ for every real number n, where k is a constant. If $g(4) = 82$, what is the value of k?

1. Set $kn = 4$, so $n = \frac{4}{k}$, and the rule gives $\frac{4}{k} + 10 = 82$, so $\frac{4}{k} = 72$.

2. Let x stand for k: type `y=4/x+10` and `y=82`.

3. Tap the intersection: (0.0556, 82). Its x-value is k.

4. Type `4/72`: it returns 0.0556, which is $\frac{1}{18}$.

Answer: $\frac{1}{18}$

2. HARD · DESMOS

A decorative fountain in a public park shoots a jet of water whose height above the ground, in meters, is modeled by

$$h(t) = -4\,(t-3)^2 + 45,$$

where t is the time, in seconds, after the jet is activated. For which values of t does the water reach a height of 29 meters?

(A) $t = 5$ only

(B) $t = 1$ and $t = 5$ ✓

(C) $t = 2$ and $t = 4$

(D) $t = 1$ only

1. Type `y=-4(x-3)^2+45` and the line `y=29`.

2. Tap the two grey dots: (1, 29) and (5, 29).

3. The height is 29 at $t = 1$ and at $t = 5$, so both values count.

Answer: (B)

3. HARD · DESMOS

The function f is defined by

$$f(x) = a\,(x-2)\,(x+6)\,,$$

where a is a constant. The graph of $y = f(x)$ in the xy-plane passes through the point $(0, -24)$. What is the value of $f(5)$?

1. Type `y=a(x-2)(x+6)` (accept the slider for a) and plot the point `(0,-24)`.

2. Slide a until the curve passes through the point: $a = 2$.

3. Type `2(5-2)(5+6)`: it returns 66.

Answer: 66

4. HARD · DESMOS

The function f is defined by $f(x) = a \cdot 2^x$, where a is a constant. If $f(3) = 40$, what is the value of $f(5)$?

(A) 64

(B) 128

(C) 160 ✓

(D) 200

1. Type `y=a*2^x` (slider a) and plot the point `(3,40)`.

2. Slide a until the curve passes through the point: $a = 5$.

3. Type `5*2^5`: it returns 160.

Answer: (C)

5. HARD · DESMOS

A ball is thrown upward, and its height above the ground, in meters, is modeled by

$$h(t) = -5\,(t-2)^2 + 30,$$

where t is the time, in seconds, after the ball is thrown. For which values of t does the ball reach a height of 10 meters?

(A) $t = 0$ and $t = 4$ ✓

(B) $t = 1$ and $t = 3$

(C) $t = 0$ only

(D) $t = 4$ only

1. Type `y=-5(x-2)^2+30` and the line `y=10`.

2. Tap the two grey dots: (0, 10) and (4, 10).

3. The height is 10 at $t = 0$ and at $t = 4$.

Answer: (A)

6. HARD · DESMOS

The function h is defined by $h(x) = (x - a)(x - b)$, where a and b are constants and $a < b$. The graph of $y = h(x)$ crosses the x-axis at $x = -3$ and $x = 7$. What is the minimum value of h?

(A) -20 (B) -10

(C) 25 **(D) -25 ✓**

1. The zeros are -3 and 7: type `y=(x+3)(x-7)`.

2. Tap the vertex (the lowest point): $(2, -25)$.

3. The minimum value of h is -25.

Answer: (D)

7. HARD · DESMOS

The function p is defined by $p(x) = a \cdot x^3$, where a is a nonzero constant. If $p(2) = 40$, what is the value of $p(-2)$?

1. Type `40/2^3`: it returns 5, so $a = 5$.

2. Type `p(x)=5x^3` and then `p(-2)`: it returns -40.

3. An odd power flips the sign, so $p(-2) = -p(2) = -40$.

Answer: -40

8. HARD · DESMOS

The function N is defined by $N(t) = k \cdot 2^{-at}$ for positive constants k and a. The table shows values of N for select values of t.

t	0	1	2
$N(t)$	200	50	12.5

What is the value of a?

1. From $N(0) = 200$ we get $k = 200$. Type `y=200*2^(-a*x)` (slider a).

2. Plot the points `(1,50)` and `(2,12.5)`.

3. Slide a until the curve passes through both points: $a = 2$.

Answer: 2

9. HARD · DESMOS

The variable z varies jointly with x and the square of y. If $z = 48$ when $x = 3$ and $y = 4$, what is the value of z when $x = 5$ and $y = 2$?

(A) 10 (B) 16

(C) 20 ✓ (D) 25

1. Write $z = kxy^2$. Type `48/(3*4^2)` : it returns $k = 1$.

2. Type `1*5*2^2` : it returns 20.

3. So $z = 20$ when $x = 5$ and $y = 2$.

Answer: (C)

10. HARD · DESMOS

The graph of $y = h(x)$ is the graph of $y = x^2$ shifted right 3 units and down 5 units. What is the value of $h(7)$?

(A) 6 **(B) 11 ✓**

(C) 14 (D) 16

1. Right 3 and down 5 gives $h(x) = (x - 3)^2 - 5$. Type `h(x)=(x-3)^2-5` .

2. Type `h(7)` : it returns 11.

3. So $h(7) = 11$.

Answer: (B)

11. HARD · DESMOS

A ball's height above the ground, in meters, t seconds after being launched from a platform is modeled by $h(t) = -5t^2 + 20t + c$, where c is the platform height. If $h(1) = 35$, what is the value of $h(3)$?

1. Type `y=-5x^2+20x+c` (slider c) and plot the point `(1,35)` .

2. Widen the slider (set its maximum to 30) and slide c until the curve passes through the point: $c = 20$.

3. Type `-5*3^2+20*3+20` : it returns 35.

Answer: 35

12. HARD · DESMOS

The function g is defined by

$$g(x) = a\,(x-3)\,(x+4)\,,$$

where a is a constant. The graph of $y = g(x)$ in the xy-plane passes through the point $(0, -36)$. What is the value of $g(6)$?

1. Type `y=a(x-3)(x+4)` (slider a) and plot the point `(0,-36)`.

2. Slide a until the curve passes through the point: $a = 3$.

3. Type `3(6-3)(6+4)` : it returns 90.

Answer: 90

13. HARD · DESMOS

A cooperative tracks its annual export tonnage with the function $f(x) = a \cdot 3^x$, where a is a constant and x is the number of years since record-keeping began. If $f(2) = 54$, what is the value of $f(4)$?

(A) 108

(B) 162

(C) 324

(D) 486 ✓

1. Type `y=a*3^x` (slider a) and plot the point `(2,54)`.

2. Slide a until the curve passes through the point: $a = 6$.

3. Type `6*3^4` : it returns 486.

Answer: (D)

14. HARD · DESMOS

The height, in meters, of a ball above the ground t seconds after being thrown is given by $h(t) = -4.9t^2 + 14.7t + 2$. Which of the following is closest to the number of seconds it takes for the ball to hit the ground ($h(t) = 0$)?

(A) 3.1 ✓

(B) 3.6

(C) 2.1

(D) 2.6

1. Type `y=-4.9x^2+14.7x+2`.

2. Tap the positive x-intercept: (3.13, 0).

3. Of the options, 3.1 is the closest.

Answer: (A)

15. HARD · DESMOS

The graph of the function $g(x) = (x-3)^2 - 7$ is the same as the graph of $y = x^2$ after which of the following transformations?

(A) Shifted left 3 and up 7

(B) Shifted right 3 and down 7 ✓

(C) Shifted right 3 and up 7

(D) Shifted left 3 and down 7

1. Type `y=(x-3)^2-7` and `y=x^2`.

2. Tap the vertex of each: $(3, -7)$ and $(0, 0)$.

3. The vertex moved 3 to the right and 7 down.

Answer: (B)

16. HARD · DESMOS

If $x = 5$ is one solution to the equation $x^2 - kx + 20 = 0$, what is the value of k?

1. Put $x = 5$ into the equation: $25 - 5k + 20 = 0$. Let x stand for k: type `y=25-5x+20` and `y=0`.

2. Tap the intersection: $(9, 0)$.

3. So $k = 9$.

Answer: 9

17. HARD · DESMOS

A company's weekly revenue, in thousands of dollars, from selling a product priced at x dollars is modeled by $R(x) = -3x^2 + 150x$. What price x maximizes the weekly revenue?

(A) 15

(B) 20

(C) 25 ✓

(D) 30

1. Type `y=-3x^2+150x`.

2. Tap the vertex: $(25, 3750)$.

3. The revenue is greatest at the price $x = 25$.

Answer: (C)

18. HARD · DESMOS

The equation $x^2 - 12x + c = 0$ has two equal (repeated) roots. What is the value of c?

1. Type `y=x^2-12x+c` (slider c).

2. Widen the slider (set its maximum to 40) and slide c until the parabola just touches the x-axis at one point.

3. That happens at $c = 36$, where the vertex is $(6, 0)$.

Answer: 36

19. HARD · DESMOS

The function f is defined by $f(x) = \dfrac{12 - x}{2 + x}$, where $x \neq -2$. As the value of x increases without bound, the value of $f(x)$ approaches which of the following?

(A) -1 ✓ (B) 1

(C) 12 (D) -2

1. Type `y=(12-x)/(2+x)` and zoom out to the right.

2. The curve flattens along a horizontal line. Type `y=-1` to confirm it hugs that line.

3. As x increases without bound, $f(x)$ approaches -1.

Answer: (A)

20. HARD · DESMOS

$y = (x - p)^2 (x + p)(x - q)$, where p and q are positive constants and $p \neq q$. How many distinct x-intercepts does the graph of this equation have in the xy-plane?

(A) 4 (B) 1

(C) 2 **(D)** 3 ✓

1. Choose $p = 1$ and $q = 2$: type `y=(x-1)^2(x+1)(x-2)`.

2. Tap the x-intercepts: -1, 1 and 2.

3. The graph has 3 distinct x-intercepts (the repeated factor adds no new intercept).

Answer: (D)

21. HARD · DESMOS

The table below shows some values of the function N.

t	0	1	2
$N(t)$	100	25	$\frac{25}{4}$

If $N(t) = k \cdot 2^{-at}$ for positive constants k and a, what is the value of a?

1. From $N(0) = 100$ we get $k = 100$. Type `y=100*2^(-a*x)` (slider a).

2. Plot the points `(1,25)` and `(2,6.25)`.

3. Slide a until the curve passes through both points: $a = 2$.

Answer: 2

22. HARD · DESMOS

The function f is defined by $f(x) = 3x + k$, where k is a constant. If $f(f(2)) = 38$, what is the value of k?

1. Type `f(x)=3x+k`: $f(2) = 6 + k$, and $f(f(2)) = 3(6 + k) + k = 18 + 4k$.

2. Type `y=18+4x` and `y=38` (letting x stand for k).

3. Tap the intersection: (5, 38), so $k = 5$.

Answer: 5

23. HARD · DESMOS

The function f is defined by $f(x) = a(x+4)(x-10)$, where a is a constant. The minimum value of f is -98. What is the value of $f(0)$?

(A) 80

(B) -98

(C) -80 ✓

(D) -40

1. Type `y=a(x+4)(x-10)` (slider a).

2. Slide a until the lowest point has $y = -98$: $a = 2$.

3. Type `2*4*(-10)`: it returns $f(0) = -80$.

Answer: (C)

24. Hard · Desmos

A theater's revenue, in dollars, from selling t tickets is $R(t) = t(320 - 10t)$. For which of the following values of t is the revenue 2,400 dollars?

(A) 18

(B) 20 ✓

(C) 26

(D) 14

1. Type `y=x(320-10x)` and the line `y=2400`.
2. Tap the two grey dots: (12, 2400) and (20, 2400).
3. Only 20 appears among the options.

Answer: (B)

25. Hard · Desmos

$y = \dfrac{5(x+3)(x-9)(x+2)}{(x-5)(x-5c)}$ In the equation above, c is a positive constant. The expression is undefined at $x = 10$ (and $x = 5$). What is the value of c?

(A) 3

(B) -3

(C) -2

(D) 2 ✓

1. The expression is undefined when $x - 5c = 0$. Type `y=x-5c` (slider c).
2. Slide c until the line crosses the x-axis at $x = 10$: $c = 2$.
3. Type `10-5*2`: it returns 0, confirming $c = 2$.

Answer: (D)

26. Hard · Desmos

The function $f(x) = ax^2 + bx + c$ has $a > 0$ and $c < 0$. Which of the following must be true?
I. f has two distinct real zeros
II. The zeros of f have opposite signs
III. The vertex of the graph of f lies in the fourth quadrant

(A) I and II only ✓

(B) I, II and III

(C) I only

(D) I and III only

1. Try an example with $a > 0$ and $c < 0$: type `y=x^2+x-6`.
2. Tap the intercepts $(-3, 0)$ and $(2, 0)$ and the vertex $(-0.5, -6.25)$.
3. Statements I and II hold (and the product of zeros is $\dfrac{c}{a} < 0$ for every such f); here the vertex is in the third quadrant, so III does not have to hold.

Answer: (A)

27. HARD · DESMOS

x	$g(x)$
-12	$-\frac{17}{4}$
-9	$-\frac{10}{3}$
3	258

The table above shows three values of x and their corresponding values of $g(x)$. If $g(x) = \frac{x^2 + a}{bx} + 2^{(cx-1)}$, what is the value of abc?

1. At $x = 3$: $258 = 2 + 2^8$, so $c = 3$. Type `2^(3*3-1)` : it returns 256, leaving $\frac{9 + a}{3b} = 2$, i.e. $9 + a = 6b$.

2. At $x = -12$ the term 2^{-37} is almost 0. Type `2^(-37)` to see it, so $\frac{144 + a}{-12b} \approx -\frac{17}{4}$, i.e. $144 + a = 51b$.

3. Type `y=6x-9` and `y=51x-144` (letting x stand for b). Tap the intersection: (3, 9), so $b = 3$ and $a = 9$.

4. Check $x = -9$: type `(81+9)/(-27)` ; it returns -3.333, which is $-\frac{10}{3}$. Then type `9*3*3` : it returns 81.

Answer: 81

28. HARD · DESMOS

The function f is defined by $f(x) = 5(3)^{x+2}$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the y-coordinate of the y-intercept of the graph of the function it defines in the xy-plane?

I. $g(x) = 45(3)^x$

II. $h(x) = 15(3)^{x+1}$

(A) Neither I nor II

(B) I only ✓

(C) II only

(D) I and II

1. Type `y=5*3^(x+2)` and tap the y-intercept: (0, 45).

2. Type `y=45*3^x` : it lies exactly on the graph, and the coefficient 45 is the intercept.

3. Type `y=15*3^(x+1)` : it also lies on the graph, but its coefficient is 15, not 45.

4. Only the first form displays the intercept.

Answer: (B)

29. Hard · Desmos

The function f is defined by $f(x) = 96(2)^{x-3}$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the y-coordinate of the y-intercept of the graph of the function it defines in the xy-plane?
I. $g(x) = 48(2)^{x-2}$
II. $h(x) = 12(2)^{x}$

(A) Neither I nor II

(B) I only

(C) II only ✓

(D) I and II

1. Type `y=96*2^(x-3)` and tap the y-intercept: (0, 12).

2. Type `y=48*2^(x-2)` : same graph, but its coefficient is 48, not 12.

3. Type `y=12*2^x` : same graph, and the coefficient 12 is the intercept.

4. Only the second form displays the intercept.

Answer: (C)

30. Hard · Desmos

The function f is defined by $f(x) = 3(4)^{x+2}$, where $x \geq 0$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?
I. $g(x) = 48(4)^{x}$
II. $h(x) = 48(2)^{2x}$

(A) I and II ✓

(B) Neither I nor II

(C) I only

(D) II only

1. Type `y=3*4^(x+2){x>=0}` and tap the lowest point, at $x = 0$: (0, 48).

2. Type `y=48*4^x` : same graph, and the coefficient 48 is the minimum.

3. Type `y=48*2^(2x)` : same graph again, with coefficient 48.

4. Both forms display the minimum.

Answer: (A)

31. HARD · DESMOS

The function f is defined by $f(x) = 6(2)^{x+3}$, where $x \geq 0$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?
I. $g(x) = 12(2)^{x+2}$
II. $h(x) = 24(2)^{x+1}$

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II ✓

1. Type `y=6*2^(x+3){x>=0}` and tap the lowest point, at $x = 0$: (0, 48).

2. Type `y=12*2^(x+2)` : same graph, but its coefficient is 12, not 48.

3. Type `y=24*2^(x+1)` : same graph, but its coefficient is 24, not 48.

4. Neither form displays the minimum.

Answer: (D)

32. HARD · DESMOS

The function $f(x) = 200(1.5)^{\frac{x}{2}}$ is defined for all x. For every increase of x by 4, the value of $f(x)$ increases by p%. What is the value of p?

1. Type `f(x)=200*1.5^(x/2)` .

2. Type `f(4)/f(0)` : it returns 2.25, the growth factor for every increase of 4.

3. Type `(2.25-1)*100` : it returns 125.

Answer: 125

33. HARD · DESMOS

The function $f(t) = 2400(0.8)^{\frac{t}{5}}$ models the value, in dollars, of a machine t years after it was bought. For every increase of t by 10, the value of $f(t)$ decreases by p%. What is the value of p?

(A) 64

(B) 20

(C) 36 ✓

(D) 40

1. Type `f(t)=2400*0.8^(t/5)` .

2. Type `f(10)/f(0)` : it returns 0.64, the factor for every increase of 10.

3. Type `(1-0.64)*100` : it returns 36, so the value decreases by 36%.

Answer: (C)

34. HARD · DESMOS

The function $N(t) = 500(2)^{\frac{t}{3}}$ models the number of bacteria in a culture t hours after the start. Which of the following gives the number of bacteria m minutes after the start?

(A) $N(m) = 500(2)^{\frac{m}{20}}$

(B) $N(m) = 500(2)^{\frac{m}{180}}$ ✓

(C) $N(m) = 500(2)^{180m}$

(D) $N(m) = 500(2)^{20m}$

1. m minutes is $\dfrac{m}{60}$ hours, so $N = 500 \cdot 2^{\frac{m}{60}/3}$. Type `N(t)=500*2^(t/3)`.

2. Test $m = 180$ minutes (3 hours): type `N(3)`; it returns 1000.

3. Type `500*2^(180/180)`: it returns 1000, a match. Type `500*2^(180/20)`: it returns 256000, so the $\dfrac{m}{20}$ form fails.

4. The exponent must be $\dfrac{m}{180}$.

Answer: (B)

35. HARD · DESMOS

The function $f(x) = 12{,}000(0.85)^x$ models the number of visitors a museum had each year, where x is the number of years since the start of 2018 and $0 \le x \le 6$. If $y = f(x)$ is graphed in the xy-plane, which of the following is the best interpretation of the y-intercept of the graph in this context?

(A) The minimum estimated number of visitors during the 6 years was 4,526.

(B) The minimum estimated number of visitors during the 6 years was 12,000.

(C) The estimated number of visitors in 2018 was 4,526.

(D) The estimated number of visitors in 2018 was 12,000. ✓

1. Type `y=12000*0.85^x{0<=x<=6}`.

2. Tap the y-intercept: (0, 12000). The input $x = 0$ is the start of 2018.

3. So 12,000 is the estimated number of visitors in 2018.

Answer: (D)

PART B · SELF-ASSESSMENT — 23 PROBLEMS

36. MEDIUM · DESMOS

The function h is defined by $h(x) = 2x^2 - 12x + 7$. What is the x-coordinate of the vertex of the graph of $y = h(x)$ in the xy-plane?

(A) 3 ✓ (B) 6

(C) -3 (D) 2

1. Type `y=2x^2-12x+7`.

2. Tap the vertex: (3, −11).

3. Check with the axis formula: type `12/(2*2)`; it returns 3. The x-coordinate is 3.

Answer: (A)

37. MEDIUM · DESMOS

The function f is defined by $f(x) = a \cdot 3^x$, where a is a constant. If $f(2) = 45$, what is the value of $f(4)$?

(A) 225 **(B) 405 ✓**

(C) 450 (D) 135

1. Type `y=a*3^x` (slider a) and plot the point `(2,45)`.

2. Slide a until the curve passes through the point: $a = 5$.

3. Type `5*3^4`: it returns 405.

Answer: (B)

38. MEDIUM · DESMOS

If $f(x) = x^2 - 4x + k$ and $f(1) = 9$, what is the value of k?

(A) 6 (B) 9

(C) 12 ✓ (D) 14

1. Put $x = 1$ into the rule: $1 - 4 + k = 9$. Let x stand for k: type `y=1-4+x` and `y=9`.

2. Tap the intersection: (12, 9).

3. So $k = 12$. Check: type `1^2-4*1+12`; it returns 9.

Answer: (C)

39. Medium · Desmos

Let $f(x) = ax^2$ and $g(x) = bx^4$ for any value of x, where a and b are positive constants. For how many values of x is $f(x) = g(x)$?

(A) Three ✓ (B) None

(C) One (D) Two

1. Pick any positive constants, for example $a = 2$ and $b = 3$: type `y=2x^2` and `y=3x^4`.

2. Tap every intersection: $(0, 0)$ and two more, one on each side of the y-axis.

3. The graphs meet three times, so $f(x) = g(x)$ for three values of x.

Answer: (A)

40. Medium · Desmos

The variable y varies inversely with x. When $x = 6$, $y = 15$. What is the value of y when $x = 10$?

(A) 25 (B) 5

(C) 6 **(D) 9 ✓**

1. Inverse variation means xy is constant: type `6*15`; it returns 90.

2. Type `y=90/x` and tap the point with $x = 10$: $(10, 9)$.

3. So $y = 9$.

Answer: (D)

41. Medium · Desmos

If $f(x) = x^2 + 1$ and $g(x) = 2x - 3$, what is the value of $f(g(2))$?

(A) 5 (B) 1

(C) 2 ✓ (D) 3

1. Type `g(x)=2x-3` and `f(x)=x^2+1`.

2. Type `f(g(2))`: Desmos works from the inside out and returns 2.

3. So $f(g(2)) = 2$.

Answer: (C)

42. Medium · No calculator needed

The function C gives the Celsius equivalent of a temperature reading x, and the function P gives the predicted crop yield, in tons, at temperature $x°$C for a greenhouse. Suppose $C(30) = 30$ and $P(30) = 18$. What is the value of $P(C(30))$?

(A) 30

(B) 18 ✓

(C) 48

(D) 540

1. Work from the inside out: $C(30) = 30$.
2. Then $P(C(30)) = P(30) = 18$.
3. Check: $C(30) = 30$ goes into P unchanged, so $P(C(30)) = P(30) = 18$.
Answer: (B)

43. Medium · Desmos

For which value of x is the function $f(x) = \dfrac{2x-1}{x^2-2x-24}$ undefined?

(A) 2

(B) 5

(C) -6

(D) -4 ✓

1. The function is undefined where the denominator is zero. Type `y=x^2-2x-24`.
2. Tap the zeros: $x = -4$ and $x = 6$.
3. Only -4 appears among the options.
Answer: (D)

44. Medium · Desmos

The function h is defined by $h(x) = 3x^2 - 24x + 10$. What is the x-coordinate of the vertex of the graph of $y = h(x)$ in the xy-plane?

(A) 4 ✓

(B) 8

(C) -4

(D) 3

1. Type `y=3x^2-24x+10`.
2. Tap the vertex: $(4, -38)$.
3. The x-coordinate is 4. Check: type `24/(2*3)`; it returns 4.
Answer: (A)

45. Medium · Desmos

During a scheduled maintenance check, the water level change in a water reservoir is modeled by $f(x) = x^3 - 5x + 12$, where $f(x)$ is the change in level, in centimeters, and x is the number of hours after maintenance begins, for $-3 \le x \le 3$. What is the value of $f(-2)$?

(A) 10

(B) 14 ✓

(C) 18

(D) 6

1. Type `f(x)=x^3-5x+12`.

2. Type `f(-2)`: it returns 14.

3. So $f(-2) = 14$.

Answer: (B)

46. Medium · Desmos

A decorative fountain shoots a stream of water whose height, in meters, is modeled by $h(t) = -5t^2 + 20t$, where t is the time in seconds after the water leaves the nozzle. What is the maximum height, in meters, reached by the stream of water?

(A) 10

(B) 15

(C) 20 ✓

(D) 25

1. Type `y=-5x^2+20x`.

2. Tap the vertex: (2, 20).

3. The maximum height is 20 meters.

Answer: (C)

47. Medium · Desmos

If $f(x) = 2x^2 - 5x + 3$, what is the value of $f(4)$?

(A) 15 ✓

(B) 17

(C) 11

(D) 13

1. Type `f(x)=2x^2-5x+3`.

2. Type `f(4)`: it returns 15.

3. So $f(4) = 15$.

Answer: (A)

48. Medium · Desmos

A farmer has 80 meters of fencing to enclose a rectangular plot, with one side against an existing wall (so fencing is needed only on the other three sides). If x represents the width of the plot (perpendicular to the wall), which function $A(x)$ models the area of the plot, in square meters?

(A) $A(x) = 80x - x^2$

(B) $A(x) = 40x - 2x^2$

(C) $A(x) = 80 - 2x^2$

(D) $A(x) = 80x - 2x^2$ ✓

1. Two widths plus the length use 80 meters, so the length is $80 - 2x$. Type `y=x(80-2x)`.

2. Type each option and hide the others in turn: `y=80x-2x^2` is drawn exactly on top of it.

3. Only that option gives the same graph as the area $x(80 - 2x)$.

Answer: (D)

49. Medium · Desmos

The function f is defined by $f(x) = (-5)(3)^x + 7$. What is the y-intercept of the graph of $y = f(x)$ in the xy-plane?

(A) $(0, -12)$

(B) $(0, -5)$

(C) $(0, 2)$ ✓

(D) $(0, 7)$

1. Type `y=-5*3^x+7`.

2. Tap the point where the curve crosses the y-axis: (0, 2).

3. The y-intercept is $(0, 2)$.

Answer: (C)

50. Medium · Desmos

x and y^2 are directly proportional. If $x = 6$ when $y^2 = 8$, what is the value of y^2 when $x = 15$?

1. Direct proportion means $x = ky^2$. Type `6/8`: it returns $k = 0.75$.

2. Type `15/0.75`: it returns 20.

3. So $y^2 = 20$ when $x = 15$.

Answer: 20

51. Medium · Desmos

Which of the following parabolas has its turning point in the second quadrant of the coordinate plane?

(A) $y = -(x+3)^2 - 1$

(B) $y = -(x+1)^2 + 2$ ✓

(C) $y = (x+2)^2 - 3$

(D) $y = (x-3)^2 - 1$

1. Type all four parabolas, one per line.
2. Tap each vertex: $(-2, -3)$, $(3, -1)$, $(-3, -1)$, $(-1, 2)$.
3. The second quadrant has $x < 0$ and $y > 0$: only $(-1, 2)$ fits.
Answer: (B)

52. Medium · Desmos

The graph of $y = 2x^2 + 12x + 11$ in the xy-plane is a parabola. What are the coordinates of its vertex?

(A) $(3, 65)$

(B) $(-3, 11)$

(C) $(-6, 11)$

(D) $(-3, -7)$ ✓

1. Type `y=2x^2+12x+11`.
2. Tap the vertex: $(-3, -7)$.
3. The vertex is $(-3, -7)$.
Answer: (D)

53. Medium · Desmos

The functions f and g are defined by $f(x) = 2x - 3$ and $g(x) = x^2 + 1$. Which expression is equivalent to $g(f(x))$?

(A) $4x^2 - 12x + 10$ ✓

(B) $4x^2 - 12x + 9$

(C) $2x^2 - 1$

(D) $4x^2 + 10$

1. Type `f(x)=2x-3` and `g(x)=x^2+1`, then `y=g(f(x))`.
2. Type each option as its own line: `y=4x^2-12x+10` lies exactly on top of the composite.
3. Only that expression gives the same graph.
Answer: (A)

54. Medium · Desmos

The function g is defined by $g(x) = \dfrac{5}{(x-2)^2 - 9}$. For which values of x is $g(x)$ undefined?

(A) $x = 2$ only

(B) $x = -1$ and $x = 5$ ✓

(C) $x = -5$ and $x = 1$

(D) $x = 11$ only

1. It is undefined where $(x-2)^2 - 9 = 0$. Type `y=(x-2)^2-9`.
2. Tap the zeros: $x = -1$ and $x = 5$.
3. So g is undefined at $x = -1$ and $x = 5$.
Answer: (B)

55. Medium · Desmos

The function $h(x) = a^x + b$, where $a > 0$ and b is a constant, passes through $(0, 10)$ and $(2, 13)$. What is the value of ab?

(A) 26

(B) 13

(C) 18 ✓

(D) 20

1. From $(0, 10)$: $1 + b = 10$, so $b = 9$. Type `y=a^x+9` (slider a) and plot `(2,13)`.
2. Slide a until the curve passes through the point: $a = 2$.
3. Type `2*9`: it returns $ab = 18$.
Answer: (C)

56. Medium · Desmos

The function $h(x) = 2(x-4)^2 - 32$ has x-intercepts at $(0, 0)$ and $(t, 0)$. What is the value of t?

1. Type `y=2(x-4)^2-32`.
2. Tap the two x-intercepts: (0, 0) and (8, 0).
3. So $t = 8$.
Answer: 8

57. Medium · Desmos

The function g is defined by $g(x) = \dfrac{2x - 4}{(x+11)(x-6)}$. If $g(a+5) = 0$, what is the value of a?

1. Type `y=(2x-4)/((x+11)(x-6))`.
2. Tap the x-intercept: (2, 0), so $g(2) = 0$ and $a + 5 = 2$.
3. Type `2-5`: it returns $a = -3$.
Answer: -3

58. HARD · DESMOS

A town's population is modeled by $P(t) = 50(3)^{\frac{t}{4}}$, where t is the number of years since 2010. According to the model, in how many years after 2010 will the population be 1,350?

1. Type `y=50*3^(x/4)` and the line `y=1350`.

2. Tap the intersection: (12, 1350).

3. The population reaches 1,350 after 12 years.

Answer: 12